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Paleomagnetism of the Precambrian Dubawnt Group

ABSTRACT

One hundred and twenty-seven oriented samples of the Dubawnt Group were collected from 32 sites (10 sites in igneous rocks and 22 in sedimentary rocks) along the shores of Baker Lake, Northwest Territories (mean location of sampling sites 64.1° N., 094.7° W.). The age of the rocks is in the range 1,720 to 1,735 m.y. Demagnetization studies, the observed consistency between igneous and sedimentary rocks, the presence of reversals, and the evidence from baked contacts indicate that the magnetization was acquired at or very soon after formation. The mean direction of magnetization, irrespective of sign, is 347°, −50° (α_{95} = 7°), and the corresponding pole is 07° N., 083° W. (α_{95} = 8°). This result, together with others recently obtained, suggests that relative to North America the pole followed a westerly track in the eastern Pacific Ocean during the middle Proterozoic. These rocks are not very different in age from the Sudbury Irruptive, and the results therefore allow a tentative reconstruction of the original attitude of the northern limb of the irruptive.

INTRODUCTION

Much paleomagnetic information is available from Precambrian rocks of the Canadian Shield with ages of 1,400 m.y. or less. Only a few other studies of the paleomagnetism of older rocks have been made (see Symons, 1971, and references therein). As a contribution to the study of these older rocks, results are now described from rocks about 1,700 m.y. old, formed soon after the Hudsonian orogeny.

GEOLOGY AND SAMPLING

The Dubawnt Group (Donaldson, 1965, 1967) comprises a lowermost red-bed sequence, a middle volcanic sequence, and an uppermost sequence of sandstone, conglomerate, dolostone, and minor volcanic rock. The rocks rest unconformably upon metamorphic rocks of the Hudsonian orogen. They are faulted, but not strongly folded, and show no evidence of regional metamorphism. The distribution of the Dubawnt Group in the vicinity of Baker Lake is shown in Figure 1.

The red-bed sequence is up to 5,000 m thick and consists of the basal South Channel Formation (mainly coarse conglomerate) and the Kazan Formation (cross-bedded arkose, siltstone, and mudstone).

These sedimentary rocks are separated by a minor unconformity from the overlying Christopher Island Formation (trachyte, andesite, and tuffaceous interflow rocks) and the Pitz Formation (largely porphyritic acid

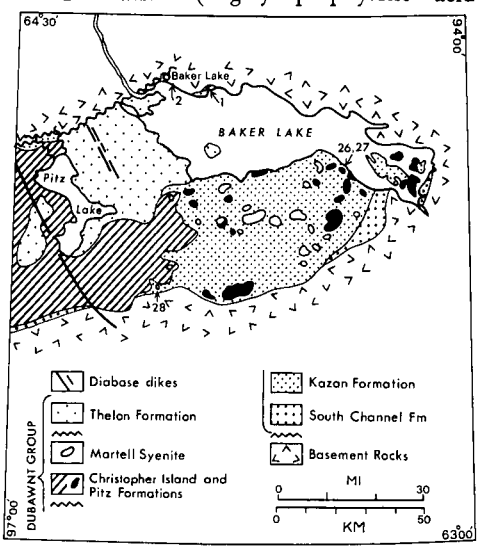


Figure 1. Baker Lake area showing distribution of lithologies of the Dubawnt Group and the locations of five sampling sites. Other sites, all near the east end of Baker Lake, are shown in Figure 2.

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flows). The volcanic sequence is at least 100 m thick. The red beds, and some of the volcanic rocks, are intruded by the Martell Syenite which is the intrusive equivalent of the Christopher Island Formation. Resting unconformably on the volcanic sequence and overlapping onto the basement rocks, are quartzose sandstone and conglomerate of the Thelon Formation which may exceed 1,000 m in thickness. The present study concerns only the Kazan Formation (20 sites in red beds), the Christopher Island Formation (nine sites: five flows, two dikes, and two red beds), and the Martell Syenite (one site).

The Hudsonian basement rocks have a mean K/Ar age of 1,735 m.y. (Stockwell, 1970). The Dubawnt igneous rocks have a mean K/Ar age (based on biotite and phlogopite) of 1,716 m.y. (five determinations), corroborated by a Rb/Sr isochron age of 1,725 ± 4 m.y. (decay constant 1.47×10^{-11} yr^{-1}; Donaldson, 1972). The age of the Kazan Formation is therefore probably in the range 1,720 to 1,735 m.y. The facts that the Kazan Formation contains volcanic detritus, that it contains iron minerals comparable in type to those occurring in the volcanic rocks (see below), and that the Christopher Island Formation contains red beds similar in aspect to those of the Kazan Formation, suggest that the Kazan red beds are only a little older than the volcanic rocks.

Most of the samples were collected from Christopher Island (Fig. 2). Cores (2.5 cm in diameter) were generally obtained, but where the rock is very hard, hand samples were collected. Samples were oriented by sun and magnetic compasses. In the laboratory two specimens were cut and measured from each core or hand sample. In the analysis, averages for each core or hand sample were used as the basic statistical unit. The remanence of specimens was measured using a Princeton Applied Research spinner magnetometer. Thermal and alternating field demagnetization was carried out using the apparatus described by Roy and others (1969, 1972, and 1973, unpub. data). The coercivities (defined as the median value of the coercive force spectrum) of natural remanent magnetization (n.r.m.), anhysteretic remanent magnetization (a.r.m.), and isothermal remanent magnetization (i.r.m.) were obtained by measuring the respective median

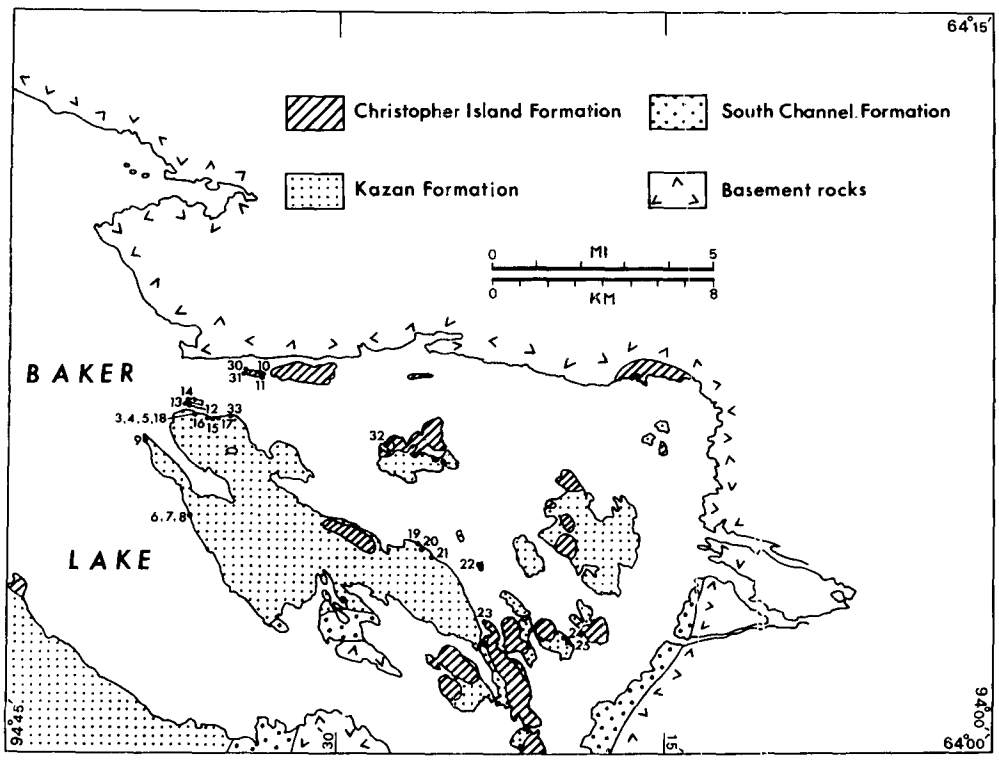


Figure 2. Sample sites at east end of Baker Lake. The largest of the islands is Christopher Island.

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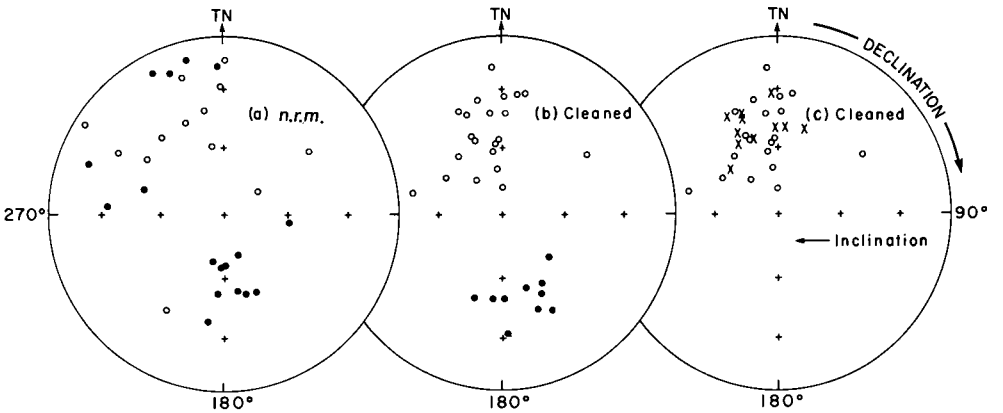


Figure 3. Mean site directions of magnetization. (a) and (b) are with respect to sign; open circles indicate negative inclination, and solid dots indicate positive inclination. In (c) the cleaned directions are plotted irrespective of sign on the upper hemisphere; open circles indicate north-seeking directions, and crosses indicate south-seeking directions.

destructive fields (m.d.f.) as described by Park and Irving (1970).

RED BEDS

The mean direction of n.r.m. at each site is plotted in Figure 3(a). Both normal and reversed polarities are present. The within-site dispersion is variable and commonly great, the mean circular standard deviation being 17° ± 12°. Pilot specimens (one or more per site) were thermally demagnetized by heating to a given temperature and cooling in zero field (±5 gammas). During demagnetization the directions moved systematically away from the present field; stable end points being reached in the range 500° to 600°C. Thermal cleaning of all specimens was carried out in this range, and the results are plotted in Figure 3(b) and listed in Table 1. After cleaning, the scatter of directions at all sites (except site 21) is improved (examples in Fig. 4), the mean standard deviation being reduced to 10° ± 6°. Thus, except for site 21, the magnetization consists of a very stable component that is unchanged in direction at high temperatures, a less stable component directed along the present field, and in some instances (for example, site 14, Fig. 4) a temporary component of random direction. An analysis of the changes in intensity upon heating shows that commonly more than half the n.r.m. is of recent origin. Site 21 has samples with both polarities that are maintained during demagnetization. In Table 1 the mean direction, irrespective of sign, for site 21 is given.

Demagnetization of i.r.m. (remanence acquired in a steady field of 10 000 Oe) by an

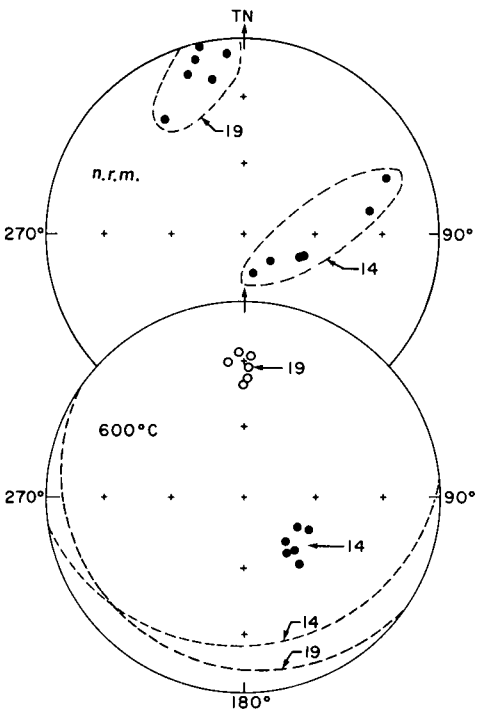


Figure 4. Examples of thermal cleaning of specimens from two sites in Kazan red beds. Open circles denote negative inclination; solid dots indicate positive inclination. The perimeter is the bedding plane, and intersections of the horizontal with the upper hemisphere are marked. All specimen directions (two per sample) are plotted. The two directions are roughly, but not exactly, 180° apart. The two sites represent spot readings of the field; the misalignment observed is within the uncertainties expected from secular variation.

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opposed steady field was carried out. The mean i.r.m. of the red beds is 0.0057 cgs units per cc, and the mean coercivity of i.r.m. is 5,230 Oe. Individual values are listed in Table 1.

Study in reflected light of polished sections from all sites shows that iron minerals occur as very fine-grained red material disseminated throughout the rock, as black crystalline particles, and as fine-grained black granular material surrounding and penetrating into the crystalline grains. The fine-grained black and red material is beyond the resolving power of the microscope. Although it cannot be identified with certainty, it is probably disseminated hematite. The black crystalline particles commonly compose about one percent of the rock and occur in the following four forms: (1) as subrounded particles 10 to 100 microns in diameter (very probably detrital) of specular hematite that almost invariably show inter-

TABLE 1. SUMMARY OF MAGNETIC PROPERTIES

Site	Rock	N	M_N	Treatment	D, I	Pole	Coercivity n.r.m.	Coercivity a.r.m.	Coercivity i.r.m.
Martell Syenite									
28	syenite	4 (3)	302.0	300 Oe	338, −51(R)	−07,104	530 (044)	215 (069)	3,890 (216)
Christopher Island Formation									
1	lava	4	531.0	1,000 Oe	056, −40(R)	−07,035	>2,000	>2,000	5,910
27	red sandstone	2	13.1	600°C	302, −59(R)	−24,131	..	..	5,000
26	lava	5	21.0	1,500 Oe	283, −46(R)	−19,151	>2,000	374	3,000
10	lava	5	366.0	1,000 Oe	355, −56(R)	−11,090	>2,000	>2,000	217
11	lava	5	146.0	1,000 Oe	352, −60(R)	−15,092	>2,000	144	264
32	red sandstone	2	5.7	600°C	001, −78(R)	−40,085	..	..	6,120
30	dike 0.5 m	6	6,930.0	400 Oe	198, +48(N)	−05,069	344	131	217
31	lava	4	41.2	150 Oe	random	..	037	212	1,600
2	dike 10 m	13	183.2	300 Oe	324, −53(R)	..	130	230	250
2	baked schist	3	4,050.0	1,000 Oe	346, −63(R)	..	217	201	361
2	combined	16	..	..	327, −55(R)	−13,111	..	..	..
9	dike 12 m	6	1,330.0	250 Oe	355, −72(M)	..	149	128	262
9	baked sediment	2	276.0	400 Oe	349, −60(R)	..	204	170	256
9	combined	8	..	..	353, −69(M)	−26,090	..	..	..
23	lapilli tuff	5	4.6	200 Oe	292, −08	..	406	267	5,090
Kazan Formation									
6	red sandstone	3	1.4	600°C	322, −70(R)	−31,111	..	..	5,500
7	red sandstone	1	1.6	600°C	357, −55(R)	−10,088	..	..	5,270
8	red sandstone	3	5.9	600°C	179, +50(N)	−05,086	..	..	5,470
18	red mudstone breccia	3	17.4	550°C	152, +38(N)	+02,112	..	..	5,560
3	red sandstone	5	5.2	550°C	186, +49(N)	−04,080	..	..	5,300
5	red sandstone	4	31.0	600°C	169, +41(N)	+01,104	..	..	5,160
4	red sandstone	5	4.1	550°C	162, +53(N)	−09,100	..	..	5,680
15	red mudstone breccia	4	26.1	600°C	177, +32(N)	+08,088	..	..	5,640
16	red sandstone	2	2.8	600°C	353, −42(R)	+01,091	..	..	5,500
17	red sandstone	3	7.2	600°C	002, −42(R)	+02,084	..	..	4,950
33	red sandstone	3	2.4	600°C	337, −37(R)	+04,106	..	..	4,870
12	red sandstone	3	14.6	550°C	150, +52(N)	−09,111	..	..	4,270
13	red sandstone	3	13.5	600°C	153, +57(N)	−05,109	..	..	4,500
14	red sandstone	3	22.6	600°C	132, +60(N)	−21,122	..	..	4,500
19	red mudstone	4	11.4	600°C	001, −34(R)	+08,084	..	..	4,940
20	red mudstone	6	15.0	550°C	007, −32(R)	+09,079	..	..	5,010
21	red mudstone	6	8.0	600°C	356, −18(M)	+16,090	..	..	5,000
22	red mudstone	3	8.7	600°C	339, −40(R)	+02,104	..	..	4,490
24	red sandstone	3	11.9	600°C	339, −52(R)	−03,103	..	..	5,850
25	red mudstone breccia	3	14.3	600°C	348, −34(R)	+07,096	..	..	6,480

Note. N is the number of separately oriented cores and samples. M_N is the intensity of natural remanent magnetization in cgs per $cm^3 \times 10^{-6}$. In the fifth column the cleaning conditions are specified. D, I are the declination and inclination of the mean site direction after cleaning. This direction is of normal (N), reversed (R), or mixed (M) polarity. The directions at sites 9 and 21 are calculated irrespective of sign. The coercivities are given in oersteds.

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growth textures ranging in scale from 10 microns down to 2 microns and less; (2) as abundant, very small (3 microns or less) grains of specularite distributed throughout the rock; (3) as irregular fillings of specular hematite in microfractures several hundred microns long and less than 10 microns wide; and (4) as small grains, up to 10 microns in size, partially replacing biotite.

VOLCANIC ROCKS

With one exception (site 31) the n.r.m. of the rocks contains a very hard component which at sites 1, 10, 11, and 26 is little changed in direction after demagnetization in an alternating field (a.f.) of 2,000 Oe. All are reversely magnetized. Most of the n.r.m. directions are highly scattered but become well grouped after partial demagnetization (Fig. 5). Demagnetization curves in the low field range are often irregular (Fig. 6). Site 31 is unstable (note the low coercivities in Table 1) and the directions are random.

The coercivities of i.r.m. are variable (Table 1). Some samples (sites 1 and 26) show two distinct stability components, one with coercivities of less than a thousand oersteds and another with coercivities of several thousand oersteds. Other samples (sites 10 and 11) show only one component with coercivities of i.r.m. between 200 and 300 Oe, yet the coercivities of n.r.m. of the same samples are at least several thousand oersteds (Table 1). Because the coercivities of titanomagnetite grains (unless the grains are very elongated) are measured in tens or hundreds of oersteds, and those of hematite in thousands of osersteds, it might be inferred that at sites 10 and 11 titanomagnetite dominates the spectrum of i.r.m. coercivities and is responsible for the unstable components that produce the high scatter of the n.r.m. directions (Fig. 5) and the irregularities in the demagnetization curves in the low field range (Fig. 6), but that hematite dominates the spectrum of n.r.m. coercivities and is responsible for the extremely stable well-grouped

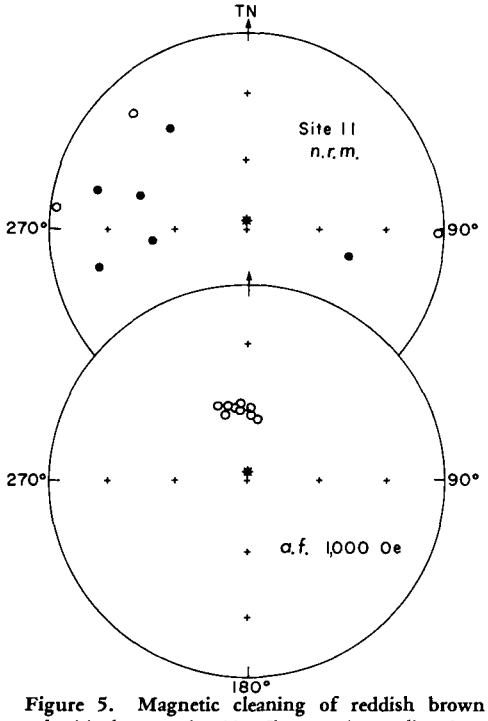


Figure 5. Magnetic cleaning of reddish brown porphyritic lava at site 11. Nine specimen directions from five cores are plotted. The present field is marked by a star.

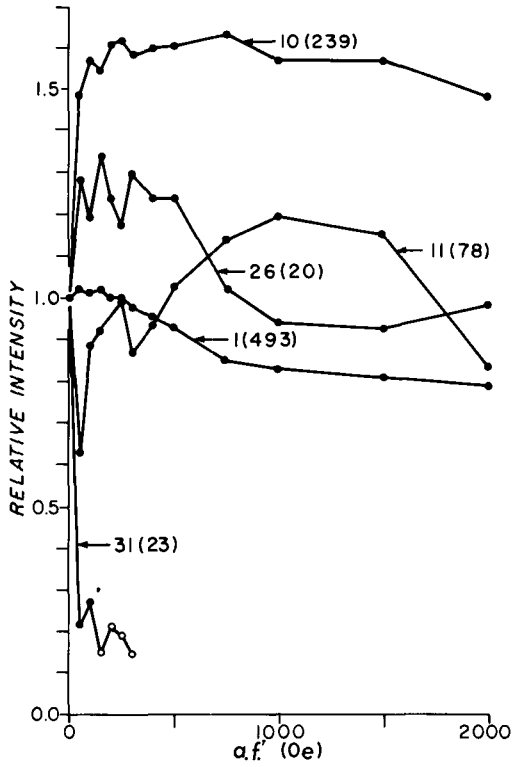


Figure 6. Alternating field (a.f.) demagnetization of lavas. The open circles (site 31) denote very large (order of 90°) and irregular direction changes. The initial (n.r.m.) intensities (cgs $cm^{-3} \times 10^{-6}$) are given in parentheses.

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directions revealed after high a.f. demagnetization (Fig. 6). On the other hand, at sites 1 and 26, hematite is apparently more abundant and dominates both the n.r.m., i.r.m., and a.r.m. spectra. This is confirmed by study of the opaque minerals. These occur as (1) abundant small crystalline grains (ranging in diameter from 100 microns to 2 microns or less) composed of intergrown specular hematite, (2) specularite partially replacing biotite, and (3) comparatively large grains (about 100 microns) consisting of a core of titanomagnetite and a rim of crystalline specular hematite. The latter grains are characteristic of the flow rocks and are particularly in evidence at sites 10 and 11 where the rocks are highly oxidized—any titanomagnetite originally present being partially or completely oxidized to specularite and ilmenite.

DIKES

Two large dikes and their baked contacts were studied. Their composition and geological relations suggest that they are feeders to flows of the Christopher Island Formation, and this is consistent with their magnetization directions. One dike (site 2) intrudes basement (Hudsonian) metamorphic rocks and is some distance away from the main outcrops of volcanic rocks. The other (site 9) intrudes Kazan sandstone. Samples from both dikes have coercivities (Table 1 and Fig. 7) characteristic of magnetite, and this is confirmed by studies of polished sections that display an abundance of brownish magnetite with intergrowth lamellae of ilmenite.

Baked rocks collected near these thick dikes are very stable, giving beautifully regular a.f. demagnetization curves (Fig. 8). Thermal demagnetization curves are square shouldered with blocking temperatures typical of magnetite (Fig. 8). Polished sections of baked red sandstone at dike 9 show abundant leucoxene and pyrite grains together with many other large opaque grains. Some of the latter are extensively corroded and consist of specularite lacking the intergrowths so common in adjacent unbaked red sandstone. Others are magnetite with cores of specular hematite. Evidently baking has reduced the hematite to magnetite, and the latter is responsible for the n.r.m.

A third, much thinner dike (site 30, Wedd Island), intrudes a sequence of brown porphyritic volcanic rock (with unstable magnetization, site 31) and sediments so that a correction for tilt can be made. The direction is very stable and has normal polarity. Abundant skeletal grains of titanomagnetite are the only iron minerals present. Intergrowths are absent and

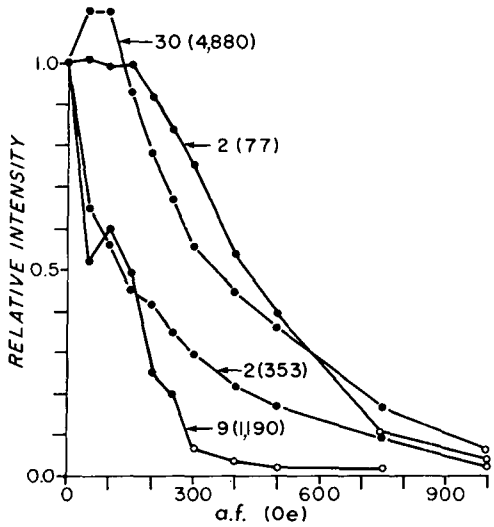


Figure 7. Alternating field demagnetization curves of dike rocks normalized to unity. Initial intensities (cgs $cm^{-3} \times 10^{-6}$) are indicated in parentheses.

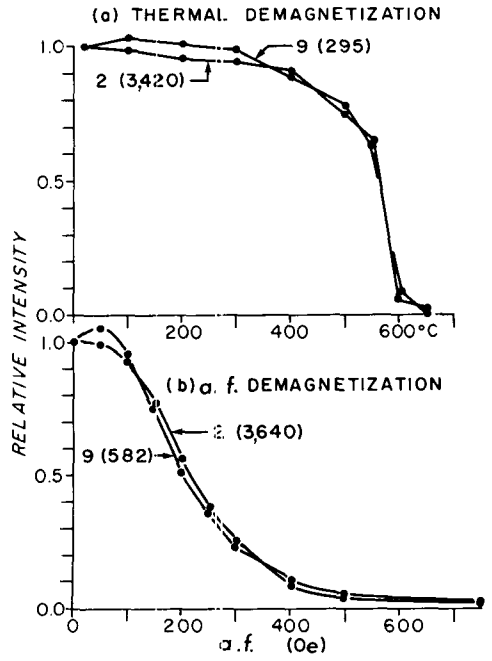


Figure 8. Thermal (a) and a.f. (b) demagnetization of rocks near dike 2 (basement greenschist 20 to 300 cm from contact) and dike 9 (baked inclusion of red sediment in dike). The initial intensities (cgs $cm^{-3} \times 10^{-6}$) are indicated in parentheses.

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the grains are absolutely fresh. The overall grain size is about 50 microns, but the snowflake texture means that the effective size is a few microns or less.

ANALYSIS OF DIRECTIONS

There is some uncertainty about the initial attitude of beds. The red beds have been slightly folded, and corrections for tilting can be estimated from regional maps. However, in an environment of rapidly accumulated beds that include alluvial fans (Donaldson, 1967), initial dips may commonly have been 5° or more. Thus, at all sedimentary sites and most igneous localities, corrections for bedding can be formally applied, although the accuracy of such corrections cannot be properly assessed on the basis of data presently available. At several igneous sites (10 and 11, for instance), field criteria for determining primary dips are lacking, but they cannot have been large, and true bedding at these sites is probably inclined at less than 10°. After correction, the between-site precision (k) increased from 14 to 17 (Table 2), but this change is not significant. We believe the directions corrected for bedding to be the more reliable, so that the first row of Table 2 (in italic) is the best summary of the data.

There are twice as many reversed sites as normal sites, but sampling may not be random with respect to time so that this apparent bias may not be significant. The mean directions of normal and reversed groups are 4° apart (Table 2, entries 4 and 5), so that, within the errors, reversal is exact. The mean inclination in igneous rocks is 12° greater than in the sedimentary rocks, but this difference is not statistically significant (rows 6 and 7 of Table 2). The general picture is one of broad agreement. The Kazan Formation and Christopher Island Formation mean directions differ by 19°, but whether this is physically real or owing to some cause such as errors in bedding corrections remains uncertain.

The mean of poles from the 30 stable sites is given in Table 2.

AGE OF MAGNETIZATION

The n.r.m. directions are commonly very scattered (Fig. 3a, Fig. 5), but after cleaning, the general grouping at 30 stable sites spanning a stratigraphic thickness of at least 2,000 m is good. The directional stability versus temperature and a.f. is often very high. In many instances (for example, site 1) the directions are essentially unchanged in a.f. as high as 2,000 Oe and temperatures of 600°C.

The magnetization of rocks baked by dikes is strong and parallel to the dikes, whereas at a distance the magnetizations are much lower (Table 1). Thermal decay curves of the baked rocks have the characteristics of thermoremanent magnetization (Fig. 8). Thus, it is almost certain that the stable magnetization of both dikes and baked rocks was acquired at the time of cooling.

In the volcanic rocks the iron minerals have been extensively oxidized, and it is the most oxidized grains (now present as hematite) that are responsible for the stable magnetization. The question therefore arises, Did this oxidation occur at the time of formation or long afterwards? On Wedd Island a thin dike (site 30) containing fresh unoxidized skeletal magnetite (an original cooling texture) cuts highly oxidized, reddish, volcanic rocks. The dike, because of its magnetization direction (Table 1), evidently belongs to the Christopher Island Formation. This indicates that the oxidation of the volcanic rocks predates the intrusion of the dike and occurred (deuterically) during the deposition of the Christopher Island Formation. Hence, it is probable that the magnetization of the volcanic rocks dates from the time of their formation.

In the red-bed sequence the mudstone breccias (sites 15, 18, and 25) are uniformly magnetized, showing that their stable magnetization was acquired subsequent to deposition. The presence of secondary infilling of microfractures by specularite in the red beds suggests that at least part of their magnetization was acquired after deposition. But the general agreement between the red beds and the igneous rocks suggests that this cannot have been later than the volcanism. The occurrence of abundant detrital specularite particles means that a component of detrital remanent magnetization (magnetization due to mechanical orientation of previously magnetized detrital particles) may be present also. Finally, the presence of mixed polarity at site 21 (see below) is a characteristic feature of magnetizations acquired during diagenesis over a period of time spanning one or more reversals (Roy and Robertson, 1968).

The general agreement among different rock types, the very high magnetic stability, the evidence from baked contacts, and the presence of reversals show that systematic components acquired long after deposition are

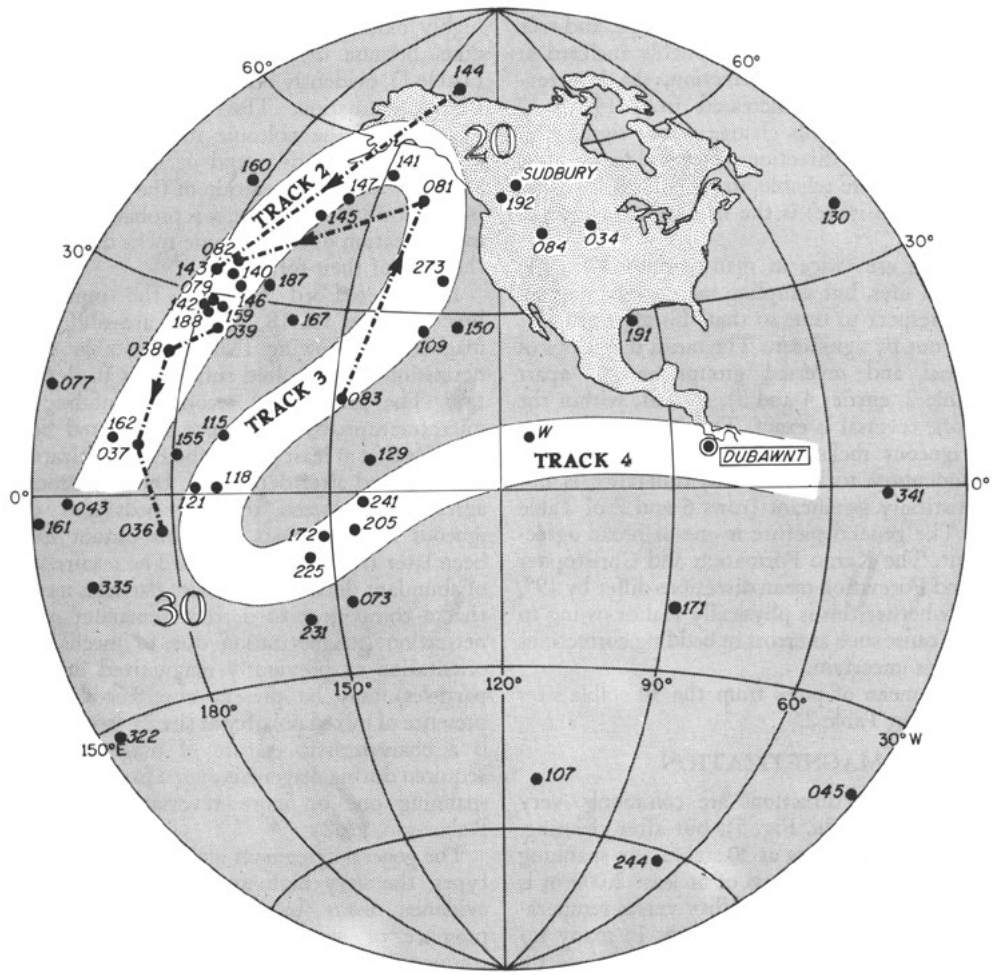
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almost certainly absent; if they were present, such consistency would not have been found. We conclude that the stable magnetization we have observed is a reliable indicator of the geomagnetic field at the time the rocks were lithified.

REVERSALS

Because of limited exposure and the possibility of repetition due to faulting, the Kazan sites cannot be reliably arranged in stratigraphic order. Within these limitations, two tentative sequences—sites 25 to 19 and sites 14 to 6—can be recognized; these sequences are based on regional dips and on a few areas of detailed sampling. The relative ages of these two sequences is not known and they may overlap. In Table 1 the sites within each sequence are arranged in approximate stratigraphic order, youngest toward the top. In the first sequence (25 to 19) the polarity is reversed, except for site 21 where both polarities occur. In the second (much thicker) sequence, both polarities occur. If the stratigraphic order is correct, then three reversals are recorded—normal to reversed, reversed to normal, and normal to reversed. Repetition due to faulting within the second sequence is possible, but this is unlikely because the reversal levels 7 to 8 (14-m separation) and 15 to 16 (9-m separation) would then have to be correlative, which in our opinion is unlikely. In the Christopher Island Formation, all sites are reversed except dike 30 which cuts some of the younger beds, so that there is evidence of at least one reversal (reversed to normal). Thus, at least two, and probably four, reversals occur. Many more may be present.

Mixed polarity occurs at site 21 in gritty red



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mudstone. Thermal demagnetization up to 650°C does not change these directions. Roy and Robertson (1968) have argued that such close intermixing of polarities in sediments could be due to an early stage detrital magnetization on which is superimposed a later chemical magnetization due to the deposition of secondary hematite. The second example of two approximately antiparallel magnetizations is in dike 9. This could be due to self-reversal or to multiple injection.

PALEOMAGNETIC POLES

The pole corresponding to the mean direction of magnetization of the Dubawnt rocks like those for the Et-Then Group (341) and the Western Channel diabase (*W* in Fig. 9) recently published (Irving and others, 1972a, 1972b) is different from pole positions previously observed from the Canadian Shield. This is shown in Figure 9 which summarizes the Precambrian poles from North America. The pole path shown is very tentative.

The grouping of poles (129, 172, 205, 241) in what is nowadays the mid-Pacific at about 1,400 m.y. is notable, and the polar loop connecting these with poles from rocks (1,250 m.y.) of the Mackenzie igneous episode (115, 118, 121), various rock units of the Keweenawan series (about 1,100 m.y.), and the Franklin diabase (162 at 675 m.y.) is recorded. The presence of such a polar loop was suggested by DuBois (1962) and has subsequently been discussed by Black (1963), Spall (1971), and Robertson and Fahrig (1971). The sharp bend that apparently occurs at the summit of this loop is referred to as hairpin 20 (Fig. 9) with an age of about 1,100 m.y. This new result from the Dubawnt Group and the Et-Then (341) and Western Channel diabase poles suggest that

Figure 9. Paleomagnetic poles for Precambrian rocks of North America. A very tentative polar wander curve (white) for the interval about 1,750 (Dubawnt) to about 700 m.y. is drawn through these poles. The bends (called "hairpins") and the tracks between them are labelled after Irving and Park (1972). The arrows indicate relative ages determined by superposition or cross-cutting relations. The poles have been compiled from Irving (1964) and from a new listing recently published in computer format (Hicken and others, 1972). The numbers refer to these listings from which the originals may be traced. Results that do not fulfill certain minimum criteria (the main criteria are that results be based on 10 or more oriented samples and that the error in the mean direction be 20° or less) are not plotted. Where two or more results are available from the same rock unit, the average pole is plotted and referenced by the lowest number. These results (the lowest numbers [italic] of which are in numerical order) with their radiometric ages in m.y. B.P. are as follows: *034*, Baraga County dikes; *036*, 074, Jacobsville Sandstone; *037*, 113, Freda Sandstone and Nonesuch Shale, 1,046; *038*, 132, Copper Harbour lavas, 1,046–1,200; *039*, 131, 148, Portage Lake lavas, Minnesota, 1,000; *043*, Umfraville Gabbro, Ontario; *045*, Tudor Gabbro, Ontario; *073*, Purcell Group; *077*, Eileen Sandstone; *079*, 264, 265, Duluth Gabbro, 1,150; *081*, 153, 269, Logan (reversed) sills, 1,000–1,100; *082*, 268, Logan (normal) dikes, 1,000–1,100; *083*, Sibley Formation, red sediments, Ontario; *084*, 139, Alona Bay lavas; *107*, Matachewan dikes, Ontario, 2,500; *109*, Abitibi dikes, 1,230; *115*, Muskox intrusion, 1,095–1,150; *118*, Mackenzie diabases, 1,250; *121*, Coppermine River Group, 1,150; *129*, Croker Island Alkali Complex, 1,475; *130*, Allard Lake anorthosite, 1,000; *140*, Mamainse Point lavas (normal), Ontario, 1,076; *141*, Mamainse Point lavas (reversed), Ontario, 1,076; *142*, Michipicoten lavas; *143*, Upper Gargantua, Ontario; *144*, Lower Gargantua, Ontario; *145*, 149, North Shore volcanics (reversed), Minnesota, 1,115; *146*, North Shore volcanics (normal), Minnesota, 1,115; *147*, Ostler Formation, Ontario; *150*, lower Keweenawan lavas, Ironwood, Michigan, 1,100; *155*, Pikes Peak Granite, Colorado, 1,030; *159*, Beaver Bay Complex, Minnesota; *160*, Cook County Gabbro, 1,000; *161*, Grenville dikes, Ontario; *162*, Franklin diabases, 675; *167*, El Paso, 953; *171*, Nipissing diabase (A), Ontario, 2,140; *172*, Sherman Granite and associated rocks, 1,410; *187*, Keweenawan intrusions, 1,100; *188*, Ancha and Sierra Ancha diabase, 1,100; *191*, Gunflint Formation, Ontario, 1,700–2,300; *192*, Hematite ore, Vermilion Range, Minnesota; *205*, Michikamau anorthosite, 1,400; *225*, Beltian Supergroup, 1,100–1,325; *231*, Southwest U.S.A., red sediments, 1,350; *241*, igneous rocks of St. Francois Mountains, 1,300–1,550; *244*, Stillwater Complex, Montana, 2,700; *273*, Nipissing diabase (C), Ontario, 1,995; *322*, Grenville metamorphic rocks; *335*, Frontenac dikes (NW), 800; *341*, Et-Then Group, 1,250–1,845; *W*, Western Channel diabase, 1,500–1,750; *Dubawnt*, 1,725; *Sudbury*, 1,712. It should be noted that the ages cited above are given to provide the reader with an *approximate* scale of time. These have been obtained by a variety of methods and are of variable reliability. This tentative polar wander path does not pass through certain poles for these reasons: 107, 171, 244 are from rocks much older than the interval under consideration; 034 is of unknown age; and 045, 130, 322 are from Grenville metamorphic rocks whose interpretation is as yet uncertain. Track 4 is roughly east-west, whereas Spall (1971) suggests that the path for this interval moves north back toward North America. This path does not fit the data in this paper and the reason for this has been suggested by Donaldson and others (1972).

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TABLE 2. ANALYSIS OF DIRECTIONS OF MAGNETIZATION

	Reference	Treatment	n	D_m	I_m	R	k	α_{95}
1. All sites	bedding	cleared	*30*	*347,*	*−50*	*28.3*	*17*	*7*
2. All sites	do.	n.r.m.	30	335,	−37	21.6	3	17
3. All sites	horizontal	cleared	30	345,	−57	27.9	14	7
4. Normal sites	bedding	do.	10	166,	+48	9.7	29	9
5. Reversed sites	do.	do.	20	348,	−51	18.6	13	9
6. Igneous rocks	do.	do.	8	350,	−59	7.4	11	18
7. Red beds	do.	do.	22	346,	−47	21.0	21	7
8. Kazan Formation	do.	do.	20	347,	−45	28.2	19	6
9. Christopher Island Formation	do.	do.	9	347,	−63	8.3	11	16
10. All poles of Table 1	—	—	*30*	*083° W., 07° N.*		—	*12*	*8*

Note. Except for entries 4 and 5, the directions are irrespective of sign. In the analysis, unit weight is given to each site (n). D_m, I_m are the declination and inclination of the mean direction. R is the length of the vector resultant. k is Fisher's index of precision. α_{95} is the radius of the circle of confidence ($P = 0.05$). Entries 1 and 10 together (italic) are considered to be the best summary of the data.

during middle Proterozoic time the pole moved back toward North America making another hairpin (30) with an approximate age of 1,350 m.y. These hairpins define the limits of tracks (numbered in Fig. 9) along which the pole moved with approximately constant direction. The hairpins mark large changes in the motion of North America relative to the pole. They correspond approximately in time to the Grenvillian and Elsonian orogenies, and it is argued elsewhere that they are causally related to them (Irving and Park, 1972; Donaldson and others, 1972).

ATTITUDE OF THE SUDBURY IRRUPTIVE

The Dubawnt rocks are approximately the same age as the Sudbury Nickel Irruptive (about 1,700 m.y.; Fairbairn and others, 1968). Hence, it is possible in principle to use the magnetization directions of the Dubawnt rocks and those previously observed from the Sudbury Irruptive to estimate the original attitude of the latter.

The present dips of the Sudbury Irruptive are generally inward toward the center of the basin, the north range values being about 30° less steep than those in the south range (Fig. 10). The original attitude is a matter of dispute; it may have been a sill-like body that was subsequently deformed, it may have been cone-shaped, or it may have had a more complex configuration. Paleomagnetic studies (Hood, 1961; Sopher, 1963; Larochelle, 1969) have shown that directions of magnetization of the north and south ranges differ (but by an amount insufficient for the sill hypothesis), and attempts have been made to determine their original attitudes relative to one another using these directions. In these procedures it is assumed that the remanent magnetization originated at the time of cooling. One procedure, followed by Hood (1961), brings the directions together by increasing the tilt of the north range by 10° and decreasing that of the south range by 30°. When this is done, the pole (Fig. 9) falls near hairpin 20 which is much younger than the Sudbury Irruptive. This might be explained by supposing that prior to its deformation the Sudbury Irruptive was remagnetized, perhaps due to heating by lavas of lower Keweenawan age (about 1,150 m.y.) that might formerly have covered the Sudbury region. This is unlikely because the results of Sopher (1963) show that the magnetization of the Sudbury Irruptive predates the Sudbury diabase dikes (pole 118, Fig. 9), which are of "Mackenzie" age (1,250 m.y.; Fahrig and Jones, 1969) and therefore are older than the lower Keweenawan lavas.

An alternative procedure that we suggest

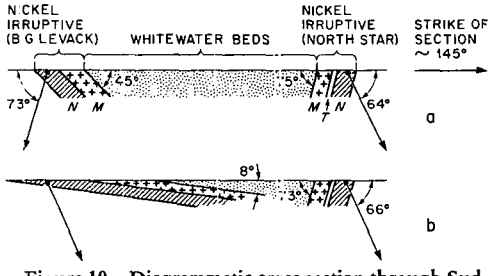


Figure 10. Diagrammatic cross section through Sudbury basin. a = present attitudes. b = reconstructed attitudes assuming remanent magnetization acquired at time of cooling and at approximately the same time on both north and south ranges.

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here is to rotate the magnetization directions of the Sudbury Irruptive into agreement with the direction observed in an essentially undisturbed sequence of rocks of approximately the same age, namely the Dubawnt Group. In this way a reconstruction of the attitudes relative to an ancient horizontal plane is made. This is done in Figure 10. Calculations using the Dubawnt direction without correction for local dip do not affect the reconstruction by more than 10°. According to this reconstruction, the Sudbury Irruptive was originally asymmetrical, the south range being essentially vertical and the north range subhorizontal. However, this result, like the earlier ones, depends for its validity on the assumption that the magnetism of the Sudbury Irruptive was acquired at the time of cooling. Presently there is no direct evidence that this is so (Larochelle, 1969). Recently, Gasparrini and Naldrett (1972) have shown that the final partition of iron and titanium in the iron minerals occurred at about 900°C on the north range and <600°C on the south range, indicating that the latter either cooled very slowly or has subsequently been reheated. This evidently is consistent with our reconstruction for the north range (Fig. 10b), but throws doubt on the reconstruction for the south range. Clearly, it will be necessary to determine more directly when the magnetization of the Sudbury Irruptive was acquired before this problem can be satisfactorily solved.

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